

AMPLIFIERS

IN

VLSI



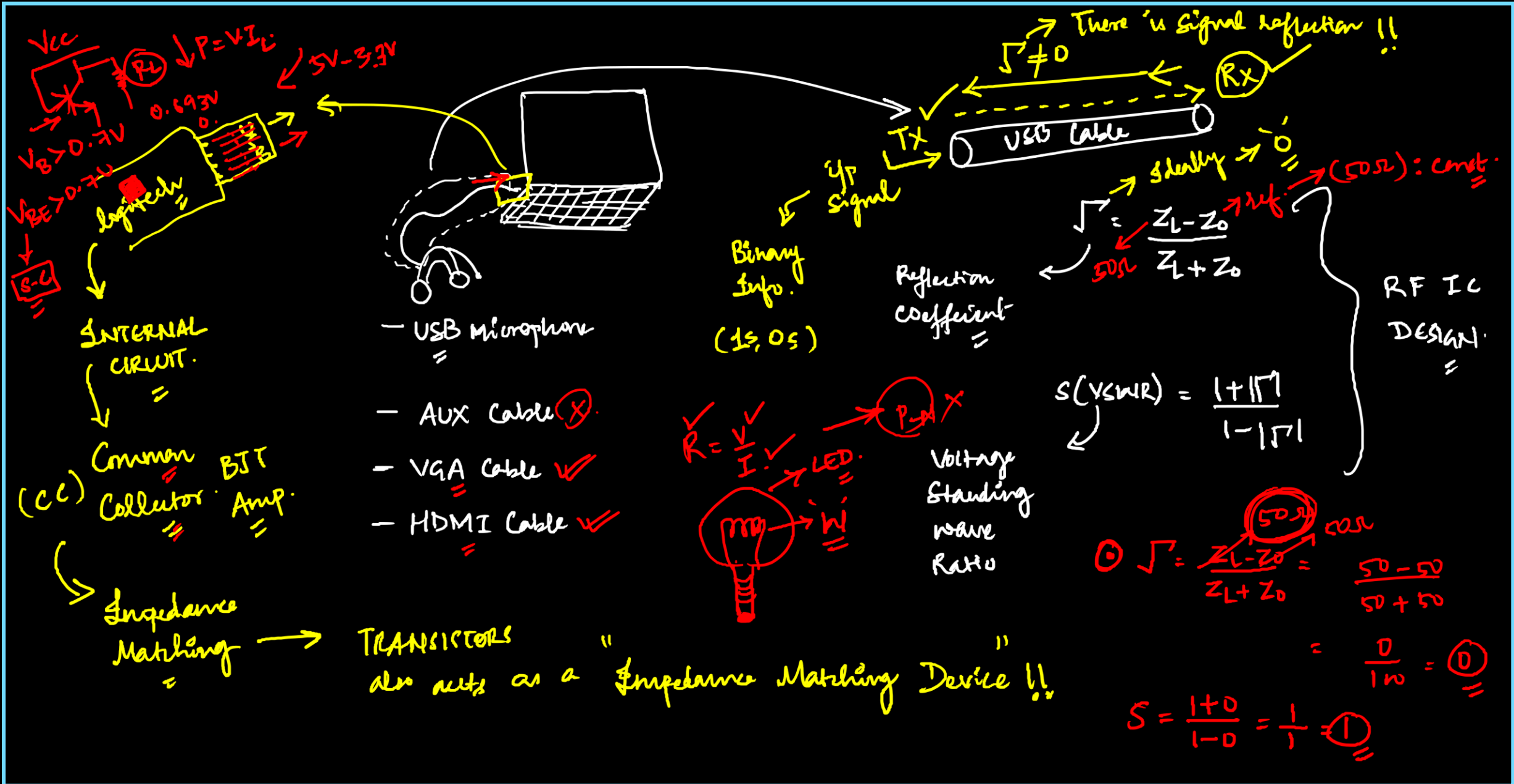
Er. Sailesh Mishra

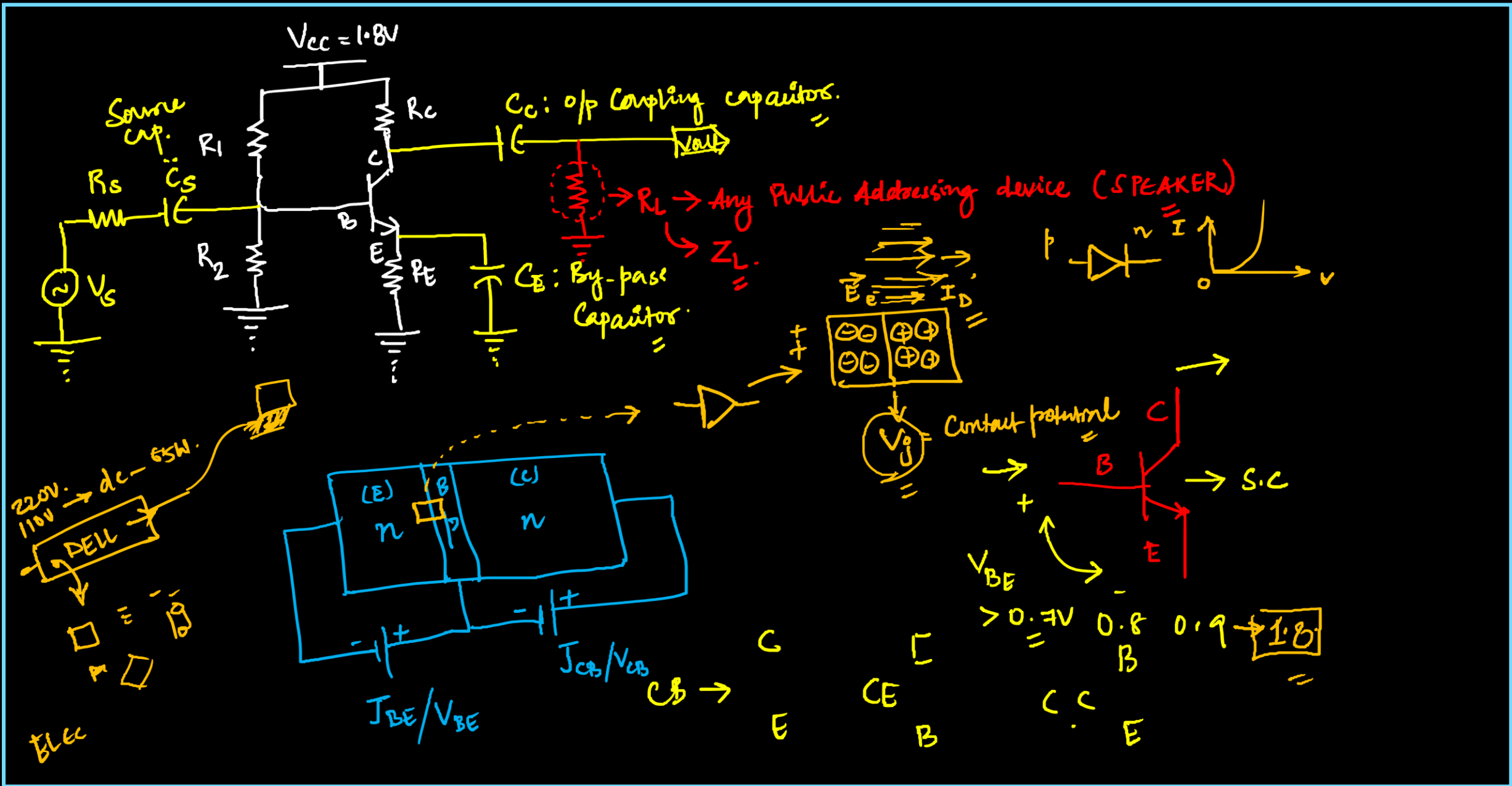
CHIP DESIGN ENGINEER

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B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI





220/- $\rightarrow$ BF ✓ (1) Min
LV ✓ (1) =
DI ✓ (1)

$\rightarrow LV \checkmark \textcircled{\downarrow}$
 $\rightarrow DI \checkmark \textcircled{\downarrow}$

$\rightarrow 9 \text{ mV} \rightarrow 90 \text{ mV} \approx Av = \frac{90}{9} = 10 \checkmark$
 $\rightarrow \textcircled{5} \text{ mV} \rightarrow 48.75 \text{ mV} \checkmark \approx \boxed{9.75} \checkmark$
 $\rightarrow 3 \text{ mV} \rightarrow 11.48 \text{ mV}$

0.25
 $\underline{\underline{=}}$

$0.1\text{ m} \rightarrow 7.6\text{ V}$.

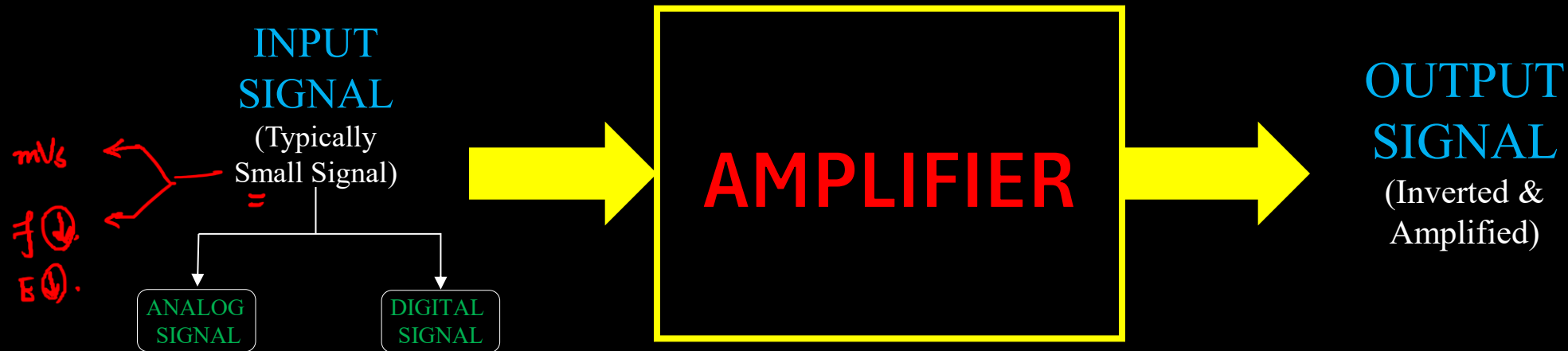
$$\Rightarrow Av = \frac{v_{out}}{v_{in}} = \frac{7.6V}{0.1m} = \frac{7.6V}{0.1 \times 10^{-3}} = \frac{7.6 \times 10^3}{0.1} \downarrow \frac{1}{10^5}$$
$$= \frac{7.6 \times 10^3 \times 10^5}{1} = 7.6 \times 10^4$$
$$= \frac{76}{10} \times 10^4 = \boxed{76000}$$

$$A_v = 76k$$

$$v_{out} = A_v v_{in} \Rightarrow$$

What is an Amplifier ??

- ❖ An “**amplifier**” is a fundamental electronic circuit or device designed to increase the magnitude of an input signal—whether voltage, current, or power—while maintaining the integrity of the original waveform within defined performance limits.
- ❖ The amplification process does not generate energy; rather, it leverages an external power source to deliver a proportionally larger output, which is achieved by employing an active device such as a BJT, JFET, MOSFET, vacuum tube, or operational amplifier, wherein the input signal serves as a control variable that modulates the controlled transfer of energy from the supply to the load.
- ❖ An amplifier is characterized by its gain, bandwidth, linearity, input/output impedance, efficiency, noise performance, and stability, each of which determines its suitability for applications ranging from low-noise instrumentation to high-power RF transmission.



❖ **Final Definition:** "An amplifier is a fundamental building block of electronics that functions as an active signal-conditioning circuit, designed to increase the amplitude of voltage, current, or power of an input signal by utilizing an "external energy source" where a semiconductor or vacuum-based active devices is employed to regulate charge carriers, thereby modulating power from the supply into a faithful but stronger reproduction of the input signal, characterized by defined gain, bandwidth, linearity, efficiency, noise performance, and stability."

VOLTAGE
GAIN ✓✓

$$A_v = \frac{V_{out}}{V_{in}}$$

→ CE
→ CB
→ CC

CURRENT
GAIN

$$A_i = \frac{I_{out}}{I_{in}}$$

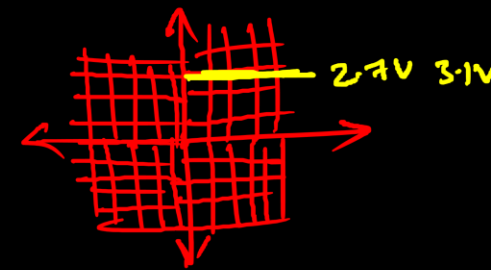
→
→
→

POWER
GAIN

$$A_p = \frac{P_{out}}{P_{in}}$$

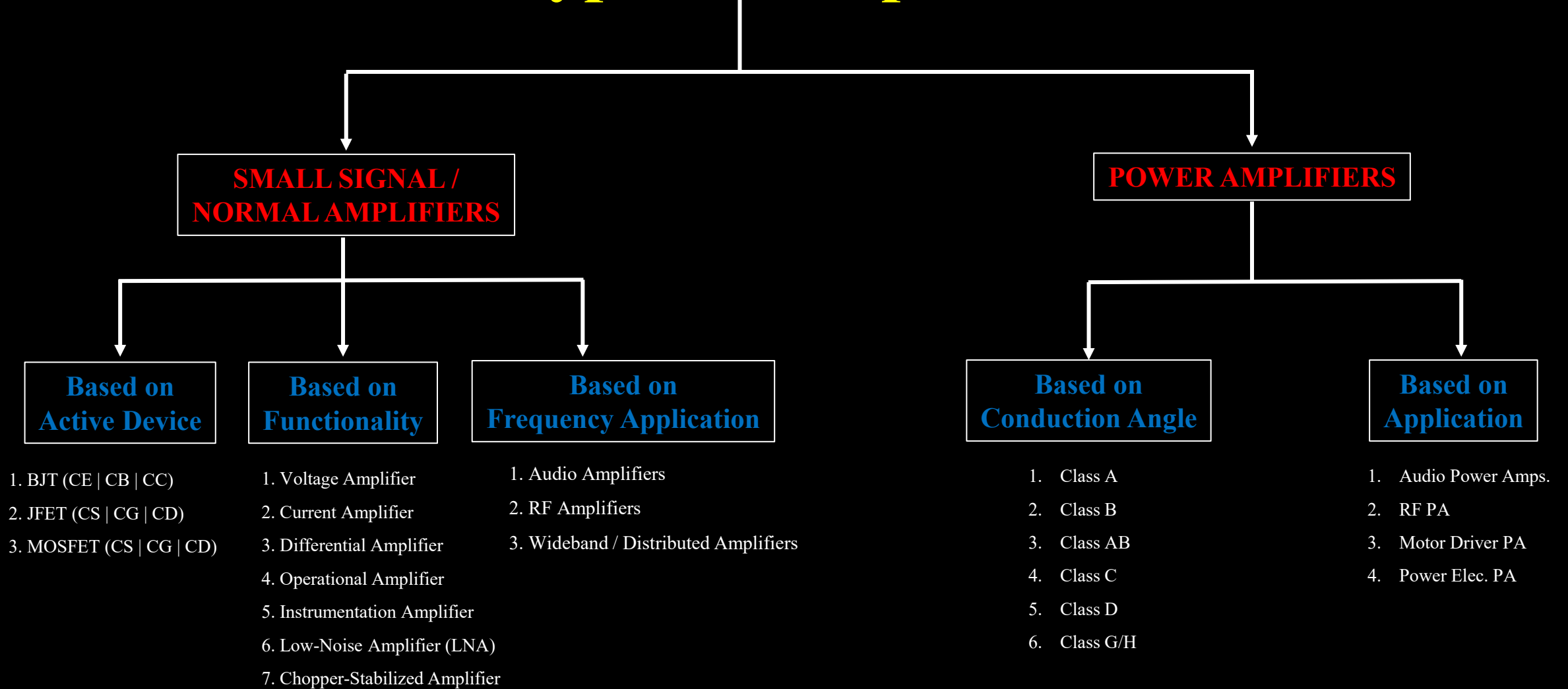
→
→
→

→ I
→ V = I · R
→ R



→ B
→ E
→ C
→ 10mV
→ 400mV
→ A·D
→ A·V = $\frac{V_{out}(V)}{V_{in}(V)}$
→ $V_{out} = A_v \times V_{in}$
→ 400mV = A · 10mV
→ A = 40

Types of Amplifier



Small Signal / Normal Amplifiers

- ❖ A small-signal amplifier is an electronic circuit designed to handle very weak input signals and make them stronger in terms of voltage or current, without necessarily delivering large amounts of power. These signals are usually in the millivolt or microamp range, coming from sensors, microphones, or antennas, and need to be boosted before further processing.
- ❖ Transistors and other active devices are naturally non-linear, meaning their input–output relationship is curved. To make them useful for amplification, they are biased at a fixed operating point, so that for small changes in input, the device responds almost linearly. In this linear region, the output becomes a scaled-up copy of the input, preserving the waveform shape but increasing its strength.
- ❖ Small-signal amplifiers are mainly about **fidelity and clarity** rather than raw power. They aim to provide :
 - i. **High gain** in voltage or current.
 - ii. **Accurate reproduction of the input waveform.**
 - iii. **Low noise**, so weak signals are not lost in interference.
 - iv. **Reasonable bandwidth**, so the signal's frequency content is preserved.
- ❖ These amplifiers are not built to drive heavy loads like speakers or motors. Instead, they prepare the signal for the **next stage** in a system, such as a power amplifier or analog-to-digital converter.
- ❖ Small-signal amplifiers are the **first and most crucial stage in many electronic systems**. They ensure that weak signals are **faithfully amplified, cleaned, and prepared** for further stages like power amplification or digital conversion. While they do not deliver high power, their importance lies in preserving **signal integrity and accuracy**, making them indispensable in communication systems, audio devices, and instrumentation.

Power Amplifiers

- ❖ A power amplifier is an electronic circuit designed not just to increase the voltage or current of a signal, but to deliver significant power to a load such as a loudspeaker, antenna, or motor. Unlike small-signal amplifiers that deal with weak signals, power amplifiers handle larger input signals and convert DC power from the supply into a high-power replica of the input waveform.
- ❖ Power amplifiers operate by controlling large currents and voltages through active devices like BJTs, MOSFETs, or vacuum tubes. The input signal provides control, while the power supply delivers the energy. The amplifier's role is to modulate this supply energy in sync with the input waveform, so that the load receives a stronger version of the signal with sufficient power to perform useful work, such as moving a speaker cone or transmitting an RF signal through an antenna.
- ❖ The design of a power amplifier **focuses more on power delivery and efficiency** rather than only signal fidelity. Important characteristics include:
 - i. **Power gain**: Ability to deliver watts of power to the load.
 - ii. **Efficiency**: Ratio of AC output power delivered to the load versus DC power drawn from the supply.
 - iii. **Linearity**: How accurately the amplifier reproduces the input waveform without distortion.
 - iv. **Load matching**: Ensuring maximum power transfer to the load (e.g., 8-ohm speaker or 50-ohm antenna).
 - v. **Thermal management**: Handling heat dissipation due to high current flow.
- ❖ Power amplifiers are the **final stage** in most electronic systems where signals need to be delivered to a practical load. Their primary goal is **power delivery with optimal efficiency**, while still maintaining acceptable fidelity. They are indispensable in applications such as **audio systems, RF transmission, motor drivers, and power electronics**, where signal strength alone is not enough and substantial energy transfer is required.